

Problem

The grid's sensors can lie — and operators act on it.

- ■ Millions of smart meters, inverters and SCADA nodes = millions of entry points
- ■ A false-data injection skews what operators see; a hijacked command skews what they do
- ■ Energy tools trust all data; security tools don't understand energy
- ■ Wrong dispatch, avoidable imports, blackouts — discovered too late

Solution

GridShield AI: predict, protect, optimize — through attacks.

- ■ Live digital twin of a solar + battery campus microgrid
- ■ AI detects when data lies, explains why, isolates the traitor device
- ■ Keeps operating on reconstructed trusted values
- ■ Re-optimizes the battery in seconds — resilience, not just alerts

How it works

Twin → AI → response, in one 1-second loop.

■ ■ Twin: deterministic physics, 5-min steps — solar, battery, grid, 6 buildings, SCADA

■ ■ AI: forecasters → 5-channel hybrid detector → classifier → explanations

■ ■ Response: isolate → reconstruct → MPC re-solve → audit log → re-admit

■ ■ FastAPI + WebSocket backend, React dashboard — runs on a laptop

Digital twin

Two data planes — the core trick.

- ■ RAW plane = exactly what SCADA sees (attacked)
- ■ TRUSTED plane = what the AI forecasts and controls with
- ■ Attacks corrupt the view, never the physics — like real FDI
- ■ Twin + attacks + detection in one loop = measurable resilience

AI

Lightweight models, calibrated decisions.

- ■ Forecast: solar 10.3 kW / demand 7.1 kW MAE — trained in <1 s
- ■ Detect: 5 evidence channels incl. control-plane consistency — the battery must obey the defender's plan
- ■ Threshold calibrated at P99.5 of clean traffic → 0.5% false positives by construction
- ■ Measured: precision 1.00 · first alert in 1 step · 5/5 attack classes

Cyber defense — live demo

One click, full chain.

■ ■ FDI_BULK launches; metered load jumps +35% — physics untouched

■ ■ ATTACK → DETECTION → EXPLANATION → DEFENSE →
RECONSTRUCTION → RE-OPTIMIZATION → RECOVERY

■ ■ Jury-readable explanations: reasons, evidence, threshold, attributions

■ ■ Same seed ⇒ identical run, every time

Results

Measured, reproducible — never invented.

- ■ Detection: precision 1.00 · FPR 0.5% · latency 1 step (all 5 attack classes)
- ■ Response: hijack exposure 25 → 5 min; battery deviation –80%
- ■ Federated FedAvg: global MAE 7.06 kW = centralized baseline, zero raw data shared
- ■ 34 tests green; every number from scripts/run_experiments.py

Impact & scalability

From campus to feeder.

- ■ Users: campuses, distribution operators, industrial parks
- ■ Value: corrupted-decision minutes avoided; battery assets protected; audit trail
- ■ Deploy: same data contract against a real historian; per-feeder detectors; ms per tick
- ■ Societal: automation that explains itself in critical infrastructure

Innovation & bonus areas

All five target areas addressed.

- ■ Digital twin with two data planes ✓
- ■ Explainable AI on every alert ✓
- ■ Live cyberattack demonstration (local, safe) ✓
- ■ Federated learning FedAvg PoC — weights only ✓
- ■ Control-plane detection channel — catches what telemetry-only detectors miss

Conclusion & roadmap

Predict. Protect. Optimize.

■ ■ Next: real SCADA ingestion (IEC 61850) · probabilistic forecasts · SHAP in production · federated detection · hardware-in-the-loop

■ ■ The detect → explain → contain → reconstruct → optimize loop is ready today

■ ■ GridShield AI — trustworthy energy automation under attack

■ IEEE SmartSecureGrid Challenge 2026 — Proof of Concept (video excluded from repo; recorded separately)